Group D · Late progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE X**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0X7-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.53** · MSIsensor score: **0.16** · AJCC edition: **4TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 33%

Model-estimated progression time

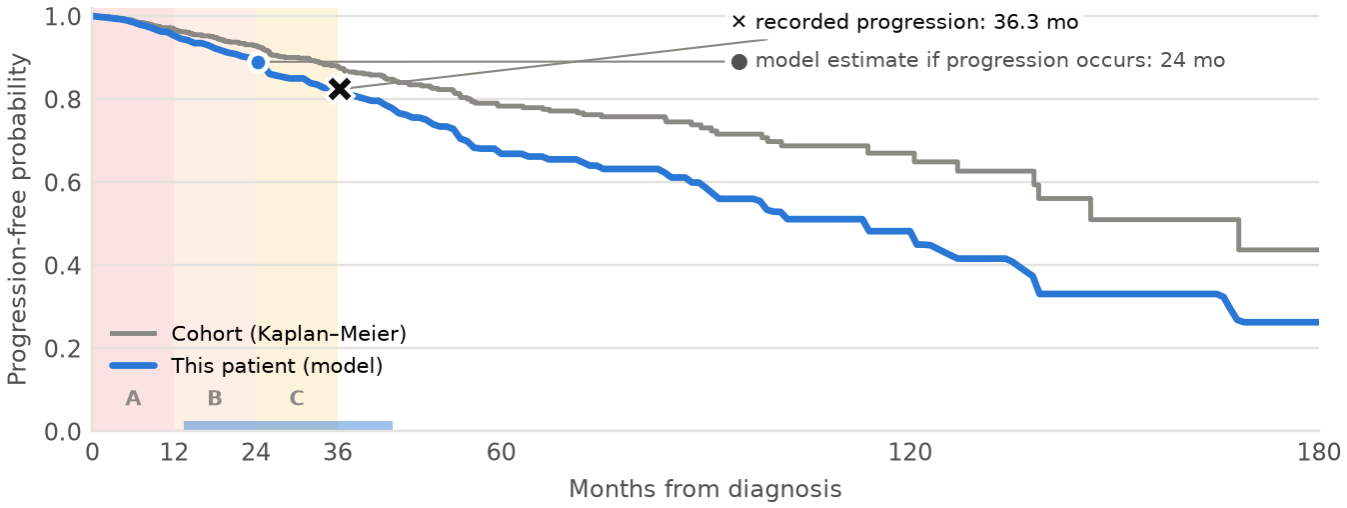
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

36.3 months

Group D · Late progression
✗ recorded window differs · |error| 12 mo



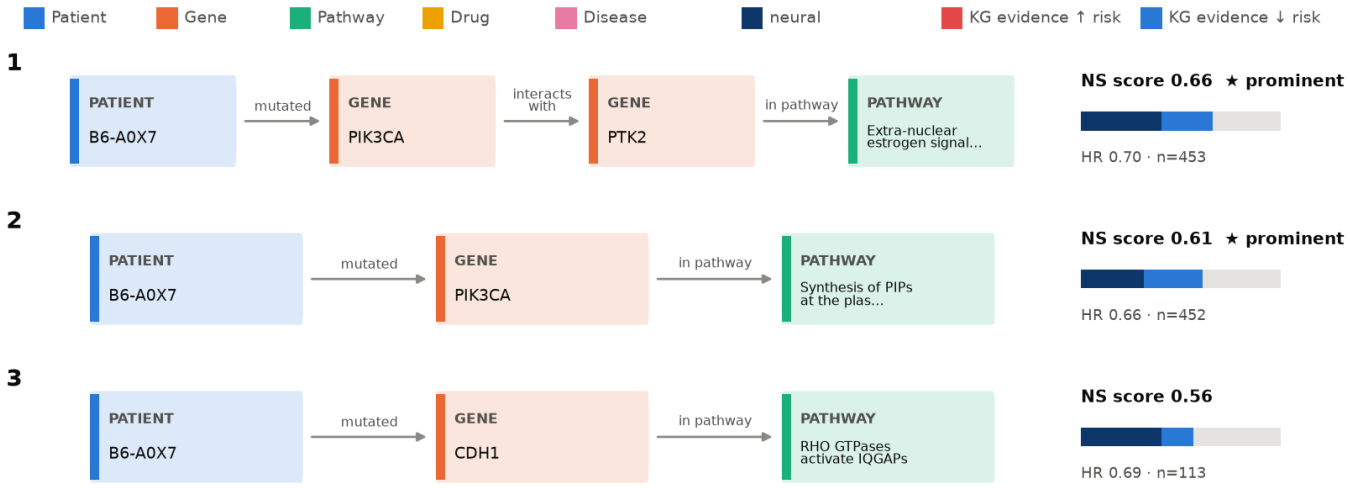
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a LumA tumour, AJCC stage X (T1, missing, MX). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Extra-nuclear estrogen signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 36.3 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

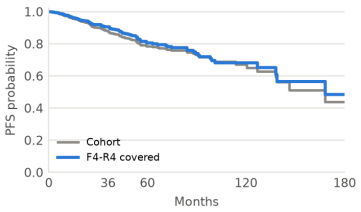


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-B6-A0X7** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **Extra-nuclear estrogen signaling**	67	↓ **slower progression** HR 0.70, z -1.9, n = 453 carriers of “Extra-nuclear estrogen signaling”	**0.66**
2 ★	**TCGA-B6-A0X7** mutated → **PIK3CA** in pathway → **Synthesis of PIPs at the plasma membrane**	52	↓ **slower progression** HR 0.66, z -2.2, n = 452 carriers of “Synthesis of PIPs at the plasma membrane”	**0.61**
3	**TCGA-B6-A0X7** mutated → **CDH1** in pathway → **RHO GTPases activate IQGAPs**	67	↓ **slower progression** HR 0.69, z -1.2, n = 113 carriers of “RHO GTPases activate IQGAPs”	**0.56**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

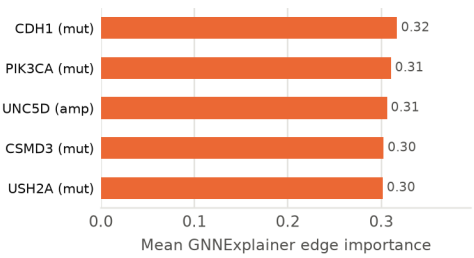
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.K675Nfs*2	17	0.32
PIK3CA · driver	Mutation	p.H1047R	70	0.31
UNC5D	High-level amplification	GISTIC ±2	1	0.31
CSMD3	Mutation	p.R615=	0	0.30
USH2A	Mutation	p.T4888M	0	0.30



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IC	90	X	LumA	0.72	no progression, followed 50.7 mo
TCGA-XX-A899	46	III	LumA	0.65	no progression, followed 15.4 mo
TCGA-E2-A576	69	I	LumA	0.60	no progression, followed 34.3 mo
TCGA-B6-A0WS	58	X	LumA	0.60	no progression, followed 97.5 mo
TCGA-EW-A1J2	50	II	LumA	0.59	no progression, followed 13.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.